- **MVC Architecture:** Models in `app/Models`, Views in `resources/views`, Controllers in `app/Http/Controllers`
- **Routing:** Defined in `routes/web.php` (browser) and `routes/api.php` (API)
- **Configuration:** Environment variables in `.env` and config files in `config/`
- **Database:** Migrations and seeders in `database/` directory
- **Artisan CLI:** Use `php artisan` for generating code, migrations, and more

> *Laravel's structure encourages **clean separation of concerns** and supports **test-driven development**, making it ideal for modern applications.*

1.8 Exercise 1: Creating Your First Laravel 12 Web Application

1.8.1 Description

In this lab, we will create a new Laravel 12 web application using Composer, set it up in Visual Studio Code, and run it using Laravel's built-in development server.

1.8.2 Objectives

By the end of this lab, you will be able to: - Install Laravel 12 using Composer - Open and configure the project in Visual Studio Code - Run the Laravel development server - View your first Laravel web app in the browser

1.8.3 Prerequisites

Make sure the following tools are installed: - PHP ≥ 8.2, https://www.php.net/downloads.php - Composer, https://getcomposer.org/ - Visual Studio Code, https://code.visualstudio.com/ - Basic terminal or command prompt knowledge

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

1.8.4 Steps